

2.4 float.h

The float header defines the minimum and maximum limits of floating-point number values.

2.4.1 Defined Values

A floating-point number is defined in the following manner:

sign value E exponent

Where *sign* is plus or minus, *value* is the value of the number, and *exponent* is the value of the exponent.

The following values are defined with the `#define` directive. These values are implementation-specific, but may not be any lower than what is given here. Note that in all instances `FLT` refers to type float, `DBL` refers to double, and `LDBL` refers to long double.

<code>FLT_ROUNDS</code>	Defines the way floating-point numbers are rounded. -1 indeterminable 0 toward zero 1 to nearest 2 toward positive infinity 3 toward negative infinity
<code>FLT_RADIX</code> 2	Defines the base (radix) representation of the exponent (i.e. base-2 is binary, base-10 is the normal decimal representation, base-16 is Hex).
<code>FLT_MANT_DIG</code> <code>DBL_MANT_DIG</code> <code>LDBL_MANT_DIG</code>	Defines the number of digits in the number (in the <code>FLT_RADIX</code> base).
<code>FLT_DIG</code> 6 <code>DBL_DIG</code> 10 <code>LDBL_DIG</code> 10	The maximum number decimal digits (base-10) that can be represented without change after rounding.
<code>FLT_MIN_EXP</code> <code>DBL_MIN_EXP</code> <code>LDBL_MIN_EXP</code>	The minimum negative integer value for an exponent in base <code>FLT_RADIX</code> .
<code>FLT_MIN_10_EXP</code> -37 <code>DBL_MIN_10_EXP</code> -37 <code>LDBL_MIN_10_EXP</code> -37	The minimum negative integer value for an exponent in base 10.
<code>FLT_MAX_EXP</code> <code>DBL_MAX_EXP</code> <code>LDBL_MAX_EXP</code>	The maximum integer value for an exponent in base <code>FLT_RADIX</code> .
<code>FLT_MAX_10_EXP</code> +37 <code>DBL_MAX_10_EXP</code> +37 <code>LDBL_MAX_10_EXP</code>	The maximum integer value for an exponent in base 10.

+37	
FLT_MAX 1E+37 DBL_MAX 1E+37 LDBL_MAX 1E+37	Maximum finite floating-point value.
FLT_EPSILON 1E-5 DBL_EPSILON 1E-9 LDBL_EPSILON 1E-9	Least significant digit representable.
FLT_MIN 1E-37 DBL_MIN 1E-37 LDBL_MIN 1E-37	Minimum floating-point value.